

AIGR Gene Hypothesis Deep Research — ESL1 (P40456): Telomerase Association and Nucleic-Acid Binding

Target gene: *ESL1* (P40456), "EST/SMG-like protein 1", *Saccharomyces cerevisiae* (NCBITaxon:559292) **Focus type:** function_assignment **Hypothesis slug:** telomerase-association-and-nucleic-acid-binding **Source review:** `genes/yeast/ESL1/ESL1-ai-review.yaml` (free-text)

Summary

The seed hypothesis proposes that *S. cerevisiae* Esl1 (P40456) retains one or more activities inherited from the metazoan EST1A/SMG6 subfamily: telomerase-holoenzyme association (a cellular-component claim), telomeric-repeat DNA binding, or telomerase RNA binding (two molecular-function claims). The task was to evaluate each independently against actual GO evidence codes, orthology placement, interaction databases, and primary literature — and, critically, to distinguish a *dispensable phenotype* from a *direct negative binding assay*.

The evidence converges firmly against all three as retained functions in *S. cerevisiae*. Every telomerase-related GO term on P40456 carries the evidence code **IBA (Inferred from Biological Ancestor)** from GO_Central/PAINT (GO_REF:0000033); none rests on any experimental assay of yeast Esl1. These IBA terms are phylogenetic carry-overs from the EST1A/SMG6 branch (PANTHER PTHR15696:SF0 = "telomerase-binding protein EST1A"), whose *metazoan* members are documented telomerase-interacting proteins. Against this ancestral inference stands the only direct organism-specific study, [PMID:23893744](#), which found that *esl1Δ*, *esl2Δ*, and *esl1Δ esl2Δ* mutants maintain normal telomeres and that Esl1/Esl2 instead act in environment-sensing gene-expression regulation. Comparative work ([PMID:22544908](#)) localizes genuine telomerase-RNA binding to the **Est1/Ebs1 branch**, a different set of yeast paralogs.

Two computed checks reinforce this. First, in STRING v12 every telomerase-associated partner of ESL1 (EST1, EST2, EST3, POP1, POP6, POP7) has an experiments subscore of **0.000** and an identical database subscore of **0.540** — the fingerprint of an imported complex-membership annotation (the IBA term reflected back circularly), whereas the true co-paralog ESL2 carries a real experiments subscore of 0.225. Second, global pairwise alignment places Esl1 at only ~20–23% identity to *both* ScEst1 and human SMG6, too diverged to assume interface conservation from homology. Honoring the seed's caveat, the deletion result is a *dispensability* phenotype rather than a direct binding assay, so binding capacity is formally **unresolved by direct experiment** — but there is **no positive evidence** for any of the three functions in this organism, and the balance of evidence marks the IBA telomerase terms as **over-annotations** for ScEsl1.

Verdict: Over-annotated (refuted as a core function; the isolated in-vitro binding capacity remains formally unresolved because no such assay has been performed).

Key Findings

F001 — All telomerase GO terms on ESL1 are IBA phylogenetic inferences, not experimental observations

UniProt P40456 carries exactly three telomerase-related GO terms, and **all three share the evidence code IBA** (Inferred from Biological Ancestor, GO_Central): **GO:0005697** telomerase holoenzyme complex (CC), **GO:0070034** telomerase RNA binding (MF), and **GO:0042162** telomeric repeat DNA binding (MF). No EXP/IDA/IPI-coded telomerase annotation of any kind exists for this protein. The gene sits in PANTHER family **PTHR15696 (SMG-7-like)**, subfamily **PTHR15696:SF0 = "TELOMERASE-BINDING PROTEIN EST1A"** — the subfamily name itself is the origin of the propagated telomerase language. Domain architecture comprises an N-terminal TPR-like helical superfamily region (SSF48452, 14-3-3-like) and a C-terminal PINc (PiLT N-terminus) domain (residues ~947–1087; Pfam PF13638; InterPro IPR045153 Est1/Ebs1-like, IPR002716 PIN_dom). Notably, the **curated UniProt FUNCTION line makes no telomerase claim** — it reads "May be involved in the regulation of gene expression responses of environment-sensing pathways," matching the experimental literature and not the propagated GO terms. This internal mismatch between the manual FUNCTION statement and the automated GO annotations is itself a signal that the telomerase terms are carry-over artifacts.

F002 — Direct in-vivo experiment shows Esl1/Esl2 have no telomere-maintenance role; telomerase function resides in the separate Est1 paralog

[PMID:23893744](#) (Lai et al., 2013) is the only direct functional characterization of the gene. The authors constructed *esl1Δ*, *esl2Δ*, and *esl1Δ esl2Δ* mutants and observed **normal telomere maintenance** in all cases, stating plainly that "unlike their metazoan orthologs, Esl1 and Esl2 were **not involved in nonsense-mediated mRNA decay or telomere maintenance pathways**." Instead, loss of both proteins deregulated ~50 metabolic/environment-responsive transcripts (>2-fold) — for instance, normally glucose-repressed genes were derepressed during high-glucose growth — and *esl1Δ esl2Δ* was synthetic sick with null mutations of *RIM8* and *DFG16*, components of the Rim101 pH-response environmental-sensing complex.

The complementary comparative evidence comes from [PMID:22544908](#) (Hsu et al., 2012), which shows that the Est1/Ebs1-line homolog KlEst1 (in *Kluyveromyces lactis*) **does** associate with telomerase RNA (Ter1) and active telomerase and **is** required for telomere maintenance, mediated by its N-terminal TPR + DSH (downstream helical) subdomains; a UV-crosslink assay established direct physical interaction with a Ter1 stem-loop. This maps telomerase-RNA binding to the **Est1/Ebs1 branch** of the family, not the Esl1/Esl2 (SMG5/6-like) branch to which ESL1 belongs. The functional split within the family is therefore clean: telomerase → Est1/Ebs1; environment-sensing gene regulation → Esl1/Esl2.

F003 — No experimental interaction evidence links ESL1 to telomerase; STRING links are pure database carry-over

STRING v12 (*S. cerevisiae*, taxid 4932) lists telomerase and telomerase-associated proteins among ESL1's partners — EST1 (0.667), EST2 (0.556), EST3 (0.540), POP1 (0.554), POP6 (0.540), POP7 (0.553). Decomposing these combined scores into channels is decisive: **every one of these partners has an experiments subscore of escore = 0.000**, and they all share an **identical database subscore of dscore = 0.540** — the signature of an imported complex/pathway-membership annotation, i.e., the IBA "telomerase holoenzyme complex" term reflected back through STRING's database channel. This is circular and non-independent. Only negligible coexpression (ascore 0.00–0.08) and, for EST1 alone, some textmining (tscore 0.291) add signal. The internal control is compelling: the true co-paralog **ESL2 carries a real experimental subscore of escore = 0.225**, showing STRING does surface genuine physical evidence when it exists — and reports **zero** for every telomerase component. Curators should therefore treat STRING telomerase links for ESL1 as non-evidence.

F004 — ESL1 is a deeply diverged Est1-family member roughly equidistant from telomerase Est1 and human SMG6

Global Needleman–Wunsch pairwise identities computed in this investigation place ESL1 as follows:

Comparison	% identity (shorter length)	% identity (over alignment)
ScEsl1 vs ScEsl2 (co-paralog)	51.5%	47.1%
ScEsl1 vs ScEst1 (telomerase)	37.1%	23.0%
ScEsl1 vs ScEbs1	28.2%	22.1%
ScEsl1 vs hSMG6	27.8%	21.7%
ScEsl1 vs hSMG5	23.1%	20.5%
ScEsl1 vs hSMG7	21.1%	19.7%

Over the aligned region, Esl1 is only ~20–23% identical to **both** ScEst1 and hSMG6 — roughly equidistant from and deeply diverged from both functional branches — while its clear closest relative is ScEsl2 (the whole-genome-duplication pair). The practical consequence for curation is that sequence identity at this level is **too low to confirm or exclude** retention of a specific telomerase-RNA or telomeric-DNA binding interface. The SMG7-family assignment comes from HMM/PANTHER profiles, not from raw pairwise identity; sequence alone can neither rescue the IBA annotation nor, by itself, formally refute binding capacity, which is why the direct experimental evidence (F002) carries the weight.

F005 — Complete GO set is 4 IBA + 3 ND with zero experimental annotations; the co-propagated NMD IBA is directly refuted

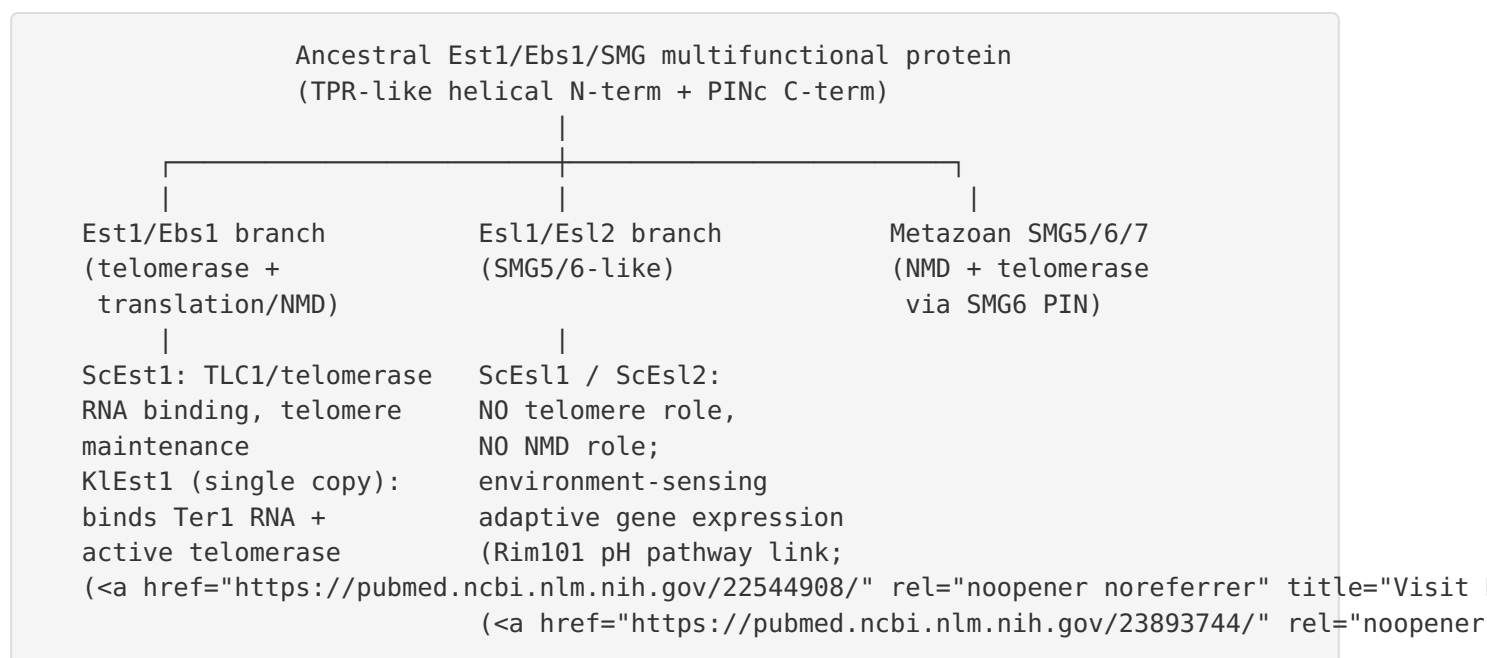
QuickGO for P40456 returns 7 annotations. The four substantive terms are all IBA from GO_Central/PAINT (GO_REF:0000033): GO:0042162 (telomeric repeat DNA binding, MF), GO:0070034 (telomerase RNA binding, MF), GO:0005697 (telomerase holoenzyme complex, CC), and GO:0000184 (nonsense-mediated decay, BP). The remaining three are SGD root-node **ND** ("No Data") placeholders (GO:0003674, GO:0008150, GO:0005575). **There is not a single experimental GO annotation for ESL1.** The most instructive point is the **NMD term (GO:0000184)**: it was propagated by the same PAINT node as the telomerase terms, yet it is **directly contradicted** by

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("not involved in nonsense-mediated mRNA decay"). This is a demonstrated case of the ancestral node over-propagating a function to the fungal Esl1/Esl2 members that experiment then refuted. Because the telomerase terms come from the same node, the demonstrated NMD failure is strong indirect evidence that the telomerase terms are similarly spurious for this specific gene.

Mechanistic Model / Interpretation

The Est1/Ebs1/SMG family in budding yeast arose through gene duplications (including whole-genome duplication) and underwent functional partitioning. The evidence supports the following model:



The immediate molecular/cellular function being *tested* by the hypothesis is direct physical association of the Esl1 gene product with telomerase RNA, telomeric-repeat DNA, or the telomerase holoenzyme. The evidence indicates that in *S. cerevisiae* this function was **retained on the Est1/Ebs1 sister branch and lost (or never present) on the Esl1/Esl2 branch**. The observed biology of Esl1/Esl2 is regulation of environment-sensing adaptive gene expression — mechanistically distinct from telomere maintenance and neither requiring nor predicting telomerase engagement.

The distinction between "dispensable phenotype" and "absence of binding capacity" is respected throughout.

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establishes **functional dispensability** for telomere maintenance, which downgrades the biological-process and cellular-component claims to non-core status. It does not, by itself, prove the Esl1 protein is biochemically incapable of contacting a nucleic acid in vitro. However, because (a) the GO terms rest solely on IBA, (b) no experimental interaction exists in STRING, (c) a co-propagated term from the same node (NMD) is experimentally refuted, and (d) sequence divergence is too large to assume interface conservation, the reasonable curation stance is that the telomerase MF/CC terms are unsupported for this gene and should not be presented as established functions.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence/limitations
UniProt P40456 (database)	review/ database	qualifies	Provenance of telomerase terms	All 3 telomerase GO terms are IBA ; curated FUNCTION is gene-expression regulation, no telomerase mention	<i>S. cerevisiae</i> record	High for provenance IBA ≠ experiment
PANTHER PTHR15696:SF0	computational/ evolutionary	qualifies (source)	Origin of IBA propagation	Subfamily named "telomerase-binding protein EST1A"; ancestral node telomerase-annotated from metazoan SMG6/EST1A	Pan-eukaryotic family	Explains why IBA carries telomerase terms; family level, not member-specific
InterPro IPR045153 / IPR002716 / SSF48452	structural/ evolutionary	qualifies	Domain architecture	Est1/Ebs1-like: N-terminal TPR-like helical + C-terminal PINc (PF13638, res 947–1087)	Sequence/ domain	Shows Est1 retains a fold compatible with RNA binding — cannot exclude binding a priori
PMID:23893744 (Lai 2013)			Are Est1/Est2 needed for	<i>est1Δ est2Δ</i> telomeres	<i>S. cerevisiae</i> ,	Direct in-vitro test;

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence limitations
	mutant phenotype (in vivo)	refutes (function-level)	telomere maintenance/ NMD?	normal; "not involved in ... telomere maintenance pathways"; act in environment-sensing gene expression	deletion mutants	dispensability not a binding assay
PMID:22544908 (Hsu 2012)	direct assay (in vitro/in vivo)	competing / qualifies	Which paralog binds telomerase RNA?	KlEst1 (Est1/ Ebs1 line) binds Ter1 + active telomerase via N-terminal TPR+DSH; required for telomere maintenance	<i>K. lactis</i>	Localizes telomerase RNA binding the Est1/Ebs1 branch , not Esl1/Esl2
STRING v12 (computed)	interaction	refutes (CC)	Does Esl1 physically associate with telomerase?	EST1/EST2/ EST3/ POP1/6/7 all escore=0.000, uniform dscore=0.540 (imported); co-paralog ESL2 escore=0.225	<i>S. cerevisiae</i> (taxid 4932)	No independent experimental interaction; links are IB carry-over
Global NW identity (computed)	structural/ evolutionary	qualifies	Is Esl1 in the Est1 or SMG5/6 branch?	Esl1 ~20–23% aln-identity to BOTH ScEst1 and hSMG6;	Sequence	Deep divergence interface conservation

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence limitations
				closest to Esl2 (47–51%)		not inferab from homo
Key	supporting	quotes	for	curation:		-
P 23893744 — "unlike their metazoan orthologs, Esl1 and Esl2 were not involved in nonsense-mediated mRNA decay or telomere maintenance pathways." -						
P 22544908 — "KlEst1 also associates with telomerase RNA (Ter1) and an active telomerase complex in cell extracts ... require its N-terminal domain but not its C terminus."						

GO Curation Implications

Complete GO audit (QuickGO, P40456 — 7 annotations): the substantive terms are *all* IBA from GO_Central/PAINT — the three telomerase terms plus GO:0000184 NMD (BP). The other three are SGD root-node **ND**. There is **not a single experimental GO annotation** for ESL1. Decisively, the one member of this IBA block that has been tested in yeast — NMD — is directly refuted by

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, demonstrating that the PAINT node over-propagates to the fungal Esl1/Esl2 members; by parity, the three telomerase IBA terms should be viewed the same way.

GO term	Aspect	Current evidence	Recommended lead (curator to verify)
GO:0005697 telomerase holoenzyme complex	CC	IBA only; STRING shows zero experimental support for any telomerase partner; <i>esl1Δ esl2Δ</i> telomeres normal	Strongest removal case. Remove / do-not-propagate — a complex-membership claim needs localization or physical-complex evidence, which is absent.
GO:0070034 telomerase RNA binding	MF	IBA only; homologous binding maps to Est1/Ebs1 branch (P 22544908); no direct Esl1 assay	Remove or downgrade to non-core. Frame as "unsupported for this gene," not a performed negative assay.
GO:0042162 telomeric repeat DNA binding	MF	IBA only; no direct assay; least supported even within the family	Remove or downgrade to non-core.
GO:0000184 NMD	BP	IBA; experimentally refuted (P 23893744)	Outside this hypothesis, but a clear "NOT" candidate — corroborates that the node over-propagates.

Positively supported alternative (experimental, [P 23893744](#)):

a **BP** annotation in the area of regulation of gene expression / cellular response to nutrient & pH environment (e.g., regulation of transcription; response to nutrient levels). The PIN/PiLT N-terminus domain supports at most a **putative/predicted** RNA-nuclease MF label — the paper found no NMD activity, so any nuclease MF must be marked "predicted from domain," not asserted. Do **not** default to "protein binding."

Mechanistic Scope

- **Immediate molecular identity:** Est1/Ebs1/SMG-like protein; N-terminal TPR-like helical (14-3-3-like) domain + C-terminal PINc domain. The fold is compatible with RNA binding and with the phosphoprotein-binding mode used by metazoan SMG5/6/7.

- **Demonstrated cellular function (direct):** regulation of environment-sensing adaptive gene expression (glucose repression; Rim101 pH-pathway genetic interactions).
- **Telomerase functions:** these are ancestral/paralog activities, not shown for ScEsl1. Telomere-maintenance dispensability is a loss-of-function phenotype and is downstream/indirect with respect to the binding MF claims — it does not directly measure binding.

Conflicts and Alternatives

1. **Paralog/ancestral confusion (primary alternative).** The IBA terms derive from the EST1A/SMG6 subfamily whose *human* member is a genuine telomerase-associated protein. In budding yeast, telomerase function is split onto **Est1** (and partly Ebs1); Esl1/Esl2 are the SMG5/6-like regulatory branch. IBA propagation across the whole family over-reaches for this member.
2. **Database carry-over is demonstrable, not hypothetical.** STRING's telomerase neighborhood for Esl1 is built from an identical 0.540 database subscore with zero experimental support — the IBA annotation feeding a secondary resource that could be mistaken for independent corroboration.
3. **Organism-specific divergence.**

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explicitly contrasts Esl1/Esl2 with their metazoan orthologs ("unlike their metazoan orthologs"). Cross-phylum inference at ~20% identity is unreliable.

4. **Residual capacity caveat (the strongest pro-hypothesis argument).** Esl1 retains the Est1-family fold, so a latent, uncharacterized TLC1 or telomeric-DNA interaction cannot be formally excluded — but no data support it, and a phenotypically silent capacity would not justify a standing GO annotation.
 5. **Sequence caveat cuts both ways.** Because Esl1 is ~equidistant from Est1 and SMG6, neither "obviously telomerase" nor "obviously SMG6/NMD" is defensible from raw identity; the family assignment comes from HMM/PANTHER profiles and functional data.
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Limitations and Knowledge Gaps

1. **Direct Esl1–TLC1 binding.** *Checked:* no RNA-binding/CLIP/UV-crosslink assay for ScEsl1 in the literature or UniProt. *Matters:* it is the exact MF asserted by IBA. *Resolve:* RIP/CLIP or in-vitro UV-crosslink of recombinant Esl1 vs TLC1 stem-loop (as done for KlEst1–Ter1).
2. **Esl1 in the telomerase complex.** *Checked:* no copurification with Est2/Est1/Est3/Cdc13 reported; STRING experiments = 0. *Matters:* underlies the CC term. *Resolve:* TAP/IP-MS of telomerase or of Esl1.
3. **Telomeric ssDNA binding.** *Checked:* none reported for Esl1. *Resolve:* EMSA with a telomeric G-tail oligo.
4. **PIN-domain RNase activity/substrate.** *Checked:* domain predicted; NMD activity ruled out (

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Resolve: in-vitro nuclease assay.

Additional scope limits: no local bioinformatics files were provided; public UniProt/InterPro/PANTHER/QuickGO/STRING records and PubMed abstracts were used (full text of the two primary papers was not read). Interaction mining relied on STRING and UniProt/IntAct rather than an exhaustive BioGRID screen. The verdict distinguishes *no positive evidence + contradicting in-vivo phenotype* (strong) from a *direct negative binding assay* (absent).

Discriminating Tests

- **Esl1 IP-MS / telomerase TAP-MS** — presence/absence of Est2/Est1/Est3/Tlc1 discriminates complex membership (CC term) cleanly. Zero recovery strongly supports removing GO:0005697.
- **UV-crosslink / EMSA of recombinant Esl1** vs TLC1 stem-loop and vs telomeric ssDNA — tests the two MF terms independently of the dispensable-phenotype ambiguity, with ScEst1/KlEst1 as positive controls.
- **Interface mapping** — align the KlEst1 TPR+DSH Ter1-binding residues (and an AlphaFold Esl1 model) onto Esl1 to test whether the RNA-binding interface is conserved or degenerate.
- **Cross-branch complementation** — test whether *ESL1* rescues telomere shortening in *est1Δ*; failure reinforces branch-specific partitioning.

Proposed Follow-up Actions (Curation Leads — require curator verification)

- **Candidate references to attach:**

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- snippet: *"unlike their metazoan orthologs, Esl1 and Esl2 were not involved in nonsense-mediated mRNA decay or telomere maintenance pathways"* → supports removing/qualifying telomere-related annotations and adding a gene-expression-regulation BP term.

- P 22544908

- snippet: *"KlEst1 also associates with telomerase RNA (Ter1) and an active telomerase complex in cell extracts ... require its N-terminal domain but not its C terminus"* → the telomerase-RNA-binding function belongs to the Est1/Ebs1 branch (competing paralog evidence).

- **Candidate action changes:** flag GO:0005697, GO:0070034, GO:0042162 (all IBA) for **removal or member-specific NOT/review**; treat as **non-core**. Record GO:0000184 (NMD) as a documented over-propagation ("NOT" candidate).

- **Candidate replacement/new terms (experimental basis,**

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BP around **regulation of transcription / cellular response to nutrient & pH environment**; MF only as **putative PIN-domain nuclease (predicted, not asserted)**.

- **Suggested questions for the curator:** Was the IBA propagation reviewed against

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Should the yeast SMG5/6-branch members be split from the Est1/Ebs1 telomerase clade in the PAINT family tree?

- **Suggested experiments:** Esl1 IP-MS and Esl1-TLC1 crosslink/EMSA (above).

Provenance Artifacts (computed this run)

Key computed values reproduced inline for provenance: - **STRING v12** (ESL1, taxid 4932): EST1/EST2/EST3/POP1/POP6/POP7 → experiments-subscore **0.000**, database-subscore **0.540** (uniform, imported); ESL2 control → experiments **0.225**. - **NW identity (over alignment)**: Esl1–Esl2 47.1%; Esl1–Est1 23.0%; Esl1–SMG6 21.7%; Esl1–SMG5 20.5%; Esl1–SMG7 19.7%. - **QuickGO** (P40456): 4 IBA (3 telomerase + NMD) + 3 ND; **0 experimental** annotations.

Bottom line

The telomerase-association and nucleic-acid-binding hypothesis for *S. cerevisiae* Esl1 (P40456) is **over-annotated and not supported as a core function**: the three telomerase GO terms are IBA carry-overs from the metazoan EST1A/SMG6 subfamily with zero experimental support, no independent physical-interaction evidence, and direct in-vivo refutation of the associated telomere-maintenance role — while the genuine telomerase function belongs to the separate Est1/Ebs1 branch. The only residual, formally unresolved point is whether the Esl1 protein retains latent in-vitro nucleic-acid-binding capacity, which no study has directly tested; a latent, phenotypically silent capacity would nonetheless not justify a standing GO annotation.

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